SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF MATHEMATICS
21MAB204T - PROBABILITY AND QUEUEING THEORY
Academic Year: 2025 - 2026 (Even Semester)

UNIT - IV : TUTORIAL SHEET – 12

S. No.	Problems
1.	At a railway station, only one train is handled at a time. The railway yard is sufficient only for two trains to wait while other is given signal to leave the station. Trains arrive at the station at an average rate of 6 per hour and the railway station can handle them on an average of 12 per hour. Assuming Poisson arrivals and exponential service distribution, find the steady state probabilities for the number of trains in the system. Also find the average waiting time of a new train coming into the yard. If the handling rate is reduced to half, what is the effect of the above results?
2.	Patients arrive at a clinic according to Poisson distribution at a rate of 60 patients per hour. The waiting room does not accommodate more than 14 patients. Investigation time per patient is exponential with mean rate of 40 per hour. (i) Determine the effective arrival rate at the clinic. (ii) What is the probability that an arriving patient will not wait? (iii) What is the expected waiting time until a patient is discharged from the clinic?
3.	A petrol pump with only one pump can accommodate 5 cars. The arrival of cars is Poisson with a mean rate of 10 per hour. The service time is exponentially distributed with a mean 2 minutes. How many cars are in the petrol pump on an average? What is the probability of a newly arriving customer finding the system full and leaving without availing service?
4.	A one person barber shop has 6 chairs to accommodate people waiting for a hair cut. Assume that customers who arrive when all the 6 chairs are full leave without entering the barber shop. Customers arrive at the average rate of 3 per hour and spend an average of 15 minutes in the barber chair. What is the probability that a customer can get directly into the barber chair upon arrival? (i) What is the expected number of customers waiting for a hair cut? (ii) How much time can a customer expect to spend in the barber shop? (iii) What fraction of potential customers are turned away?
5.	A drive in car wash facility with single server has a capacity to hold on its grounds a maximum of 10 cars (including the one in wash). The company has found its arrivals to be Poisson with a mean rate of 20 cars/hour and its service time to be exponential with mean of 12 minutes. What is the average number of cars lost to the firm every 10 hour day as a result of its capacity limitations?
6.	Consider a single server queueing system with Poisson input exponential service times. Suppose the mean arrival rate is 3 calling units per hour, the expected service time is 0.25 hours and the maximum permissible number calling units in the system is two. Find the steady state probability distribution of the number of calling units in the system and the expected number of calling units in the system.